

Artificial Intelligence

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Lecture 6

First-Order Logic

- First-order logic
- FOL Syntax and Semantics
- Quantifiers
- Connections between $\forall$ and $\exists$
- First-order logic for reflex agent

Recall: Problems of the propositional agent

1. There are too many propositions to handle.

For example, The simple rule “Don’t go forward if the Wumpus is in front of the agent” can be stated in propositional logic by 64 rules, i.e., 16 squares x 4 directions = 64.

2. Since the world can change configuration at each time step, we must specify **time** in the inference rules.

$$A_{1,1}^0 \wedge East_A^0 \wedge \neg W_{2,1} \Rightarrow Forward^0$$

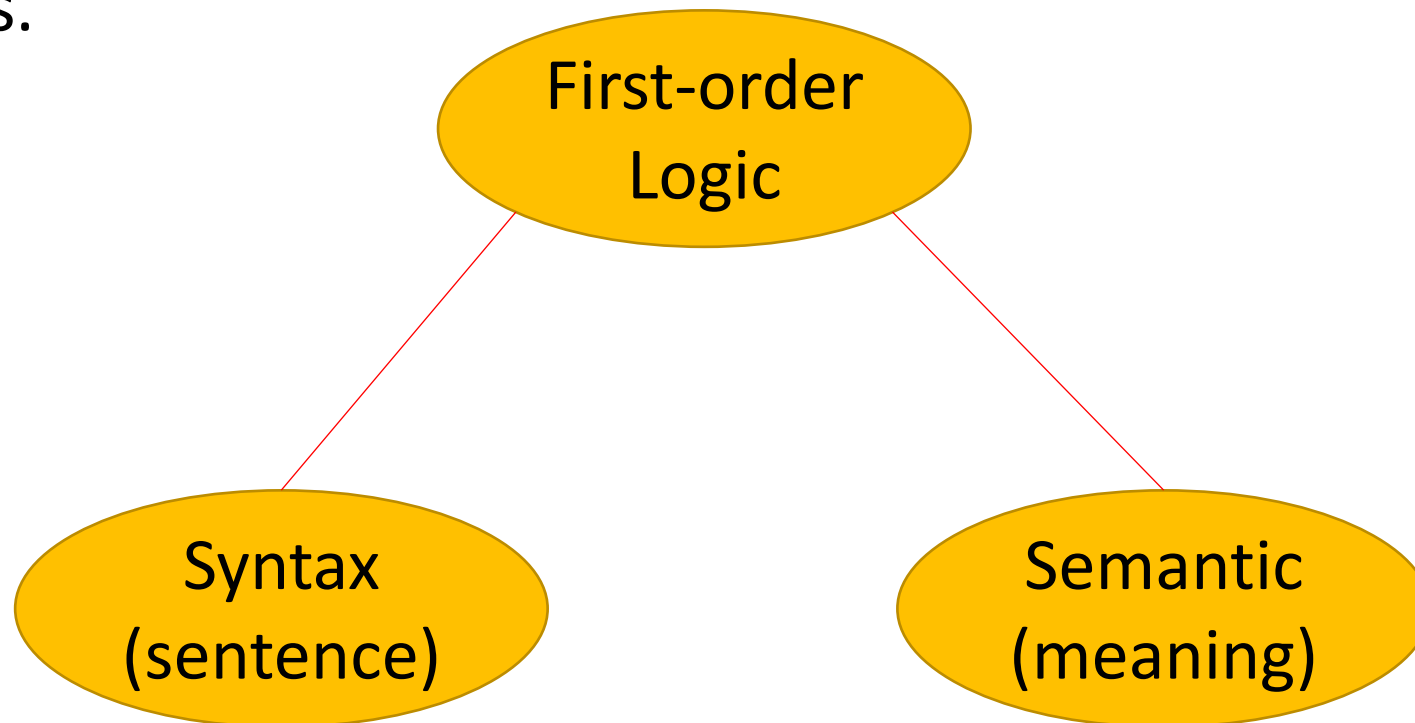
$$A_{1,1}^6 \wedge East_A^6 \wedge \neg W_{2,1} \Rightarrow TurnLeft^6$$

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2	3,2	4,2
1,1 A OK	2,1 OK	3,1	4,1

First-Order Logic

First-Order Logic

- First-order logic can express the properties of entire collections of objects rather than having to enumerate the objects by name.
- Similar to other logics, First-order logic comprises syntax and semantics.



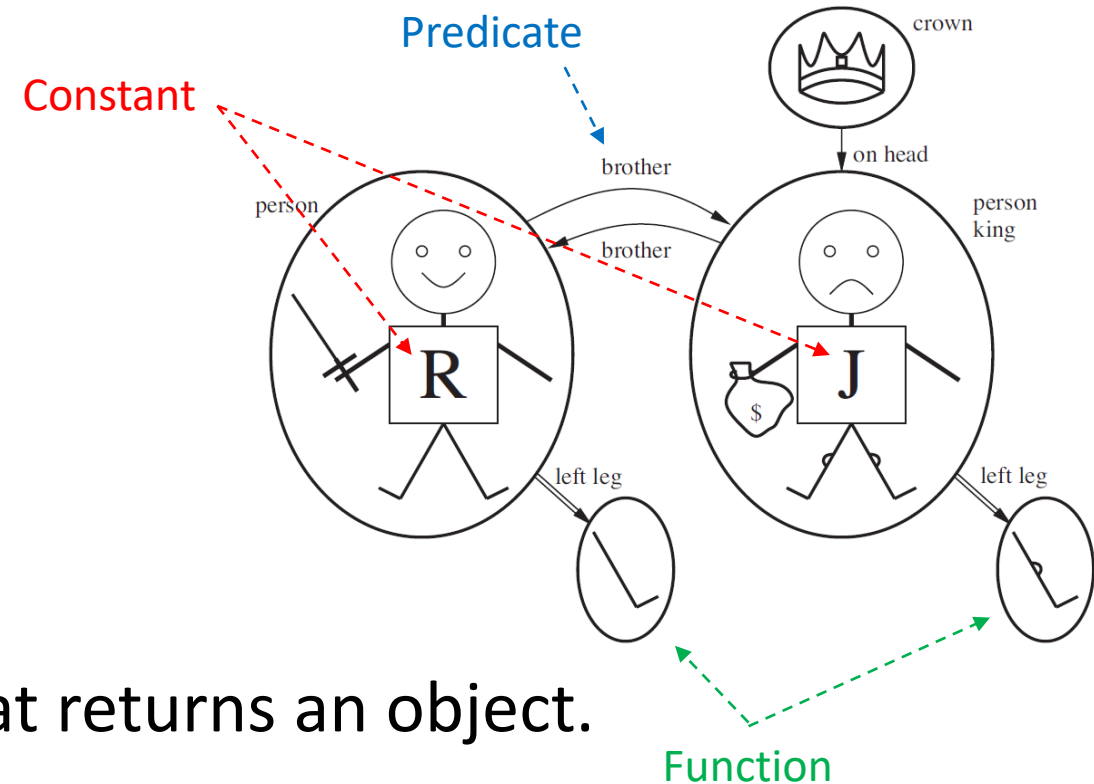
Syntax (Symbol, Atomic sentence, Complex sentence)

- **Symbol** consists of constant, predicate, and function.

- **Constant** symbol: object
Ex: Richard, John, etc.

- **Predicate** symbol: relation
Ex: Brother, Friend, etc.

- **Function** symbol: a function that returns an object.
Ex: LeftLeg, etc.



Syntax

- **Atomic sentence** represents a relationship between objects:- formed from a predicate symbol followed by a parenthesized list of objects.

For example,

Brother(Richard, John), Married(FatherOf(Richard), MotherOf(John))

- **Complex sentence**:- logical connectives of atomic sentences.

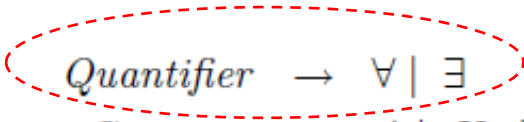
Brother(Richard, John) $\wedge$ Brother(John, Richard),

$\neg$ King(Richard) $\Rightarrow$ King(John)

The syntax of first-order logic.

$$\begin{aligned} \textit{Sentence} &\rightarrow \textit{AtomicSentence} \mid \textit{ComplexSentence} \\ \textit{AtomicSentence} &\rightarrow \textit{Predicate} \mid \textit{Predicate}(\textit{Term}, \dots) \mid \textit{Term} = \textit{Term} \\ \textit{ComplexSentence} &\rightarrow (\textit{Sentence}) \mid [\textit{Sentence}] \\ &\mid \neg \textit{Sentence} \\ &\mid \textit{Sentence} \wedge \textit{Sentence} \\ &\mid \textit{Sentence} \vee \textit{Sentence} \\ &\mid \textit{Sentence} \Rightarrow \textit{Sentence} \\ &\mid \textit{Sentence} \Leftrightarrow \textit{Sentence} \\ &\mid \textit{Quantifier} \textit{Variable}, \dots \textit{Sentence} \\ \\ \textit{Term} &\rightarrow \textit{Function}(\textit{Term}, \dots) \\ &\mid \textit{Constant} \\ &\mid \textit{Variable} \end{aligned}$$

These 2 quantifiers significantly reduce the number of rules in FOL.


$$\begin{aligned} \textit{Quantifier} &\rightarrow \forall \mid \exists \\ \textit{Constant} &\rightarrow A \mid X_1 \mid \textit{John} \mid \dots \\ \textit{Variable} &\rightarrow a \mid x \mid s \mid \dots \\ \textit{Predicate} &\rightarrow \textit{True} \mid \textit{False} \mid \textit{After} \mid \textit{Loves} \mid \textit{Raining} \mid \dots \\ \textit{Function} &\rightarrow \textit{Mother} \mid \textit{LeftLeg} \mid \dots \end{aligned}$$

OPERATOR PRECEDENCE : $\neg, =, \wedge, \vee, \Rightarrow, \Leftrightarrow$

Quantifier: express the properties of entire collections of objects, rather than having to enumerate the objects by name.

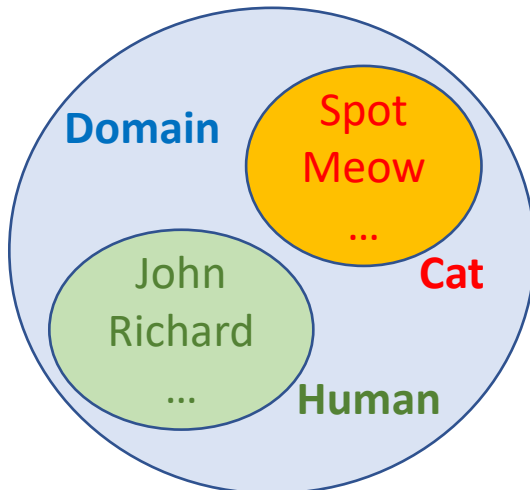
1. Universal Quantification ($\forall$, *For All*)

$$\forall_x \text{Cat}(x) \Rightarrow \text{Mammal}(x)$$

For all object x , if x is a cat then x is mammal.

If there is no any cat in our domain,
it is fine, i.e., the sentence is still true.

Problem: If we represent “All cats are mammal” by $\forall_x \text{Cat}(x) \wedge \text{Mammal}(x)$,
we will not capture what we need since the representing sentence
means the following:



$$\begin{aligned} & [\text{Cat}(\text{Spot}) \wedge \text{Mammal}(\text{Spot})] \wedge \\ & [\text{Cat}(\text{Meow}) \wedge \text{Mammal}(\text{Meow})] \wedge \\ & [\text{Cat}(\text{John}) \wedge \text{Mammal}(\text{John})] \wedge \\ & [\text{Cat}(\text{Richard}) \wedge \text{Mammal}(\text{Richard})] \wedge \\ & \dots \quad \dots \quad \dots \end{aligned}$$

Too strong sentence
(Always false)

2. Existential Quantification ($\exists$, *For Some*)

$$\exists_x \text{Sister}(x, \text{spot}) \wedge \text{Cat}(x)$$

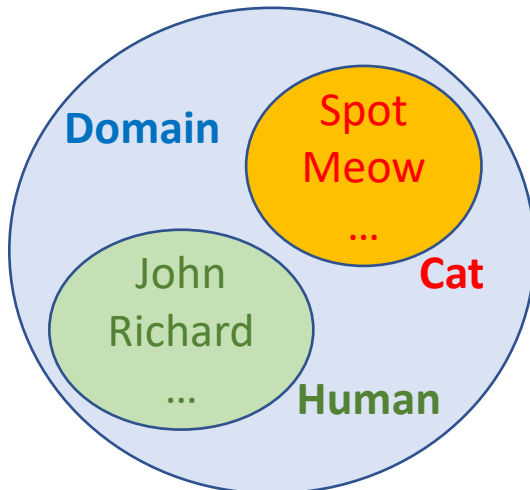
There is a sister of spot who is a cat.

If there is no sister of spot who is a cat, this sentence must be false.

Problem: If we represent “There is a sister of spot who is a cat” by

$$\exists_x \text{Sister}(x, \text{spot}) \Rightarrow \text{Cat}(x),$$

we will not capture what we need since the representing sentence means the following:



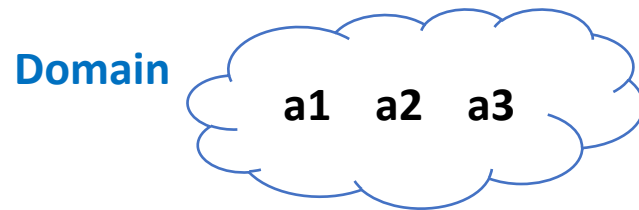
$$\begin{aligned} &[\text{Sister}(\text{Spot}, \text{Spot}) \Rightarrow \text{Cat}(\text{Spot})] \vee \\ &[\text{Sister}(\text{Meow}, \text{Spot}) \Rightarrow \text{Cat}(\text{Meow})] \vee \\ &[\text{Sister}(\text{John}, \text{Spot}) \Rightarrow \text{Cat}(\text{John})] \vee \\ &[\text{Sister}(\text{Richard}, \text{Spot}) \Rightarrow \text{Cat}(\text{Richard})] \vee \\ &\dots \quad \dots \quad \dots \end{aligned}$$

Too weak sentence
(Always true)

- Example: “Some cats are intelligent”.

$$\exists_x Cat(x) \Rightarrow Intel(x)$$

What’s wrong with the above FOL sentence ?



Name	Cat / Dog ?	Intelligent
a1	Cat	No
a2	Cat	No
a3	Dog	Yes

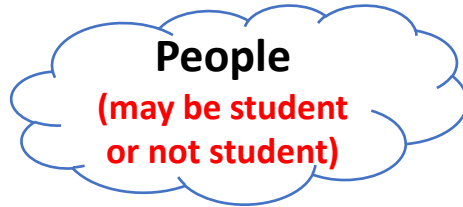
$$\underbrace{[Cat(a1) \Rightarrow Intel(a1)] \vee [Cat(a2) \Rightarrow Intel(a2)]}_{\text{False}} \vee \underbrace{[Cat(a3) \Rightarrow Intel(a3)]}_{\text{True}}$$

True

This FOL sentence always true even though there is no cat that is intelligent.

- Example: “Every student in this class has visited Africa or America”.

Domain



student(x) : x is student in this class

vaf(x) : x has visited Africa

vam(x) : x has visited America

Find FOL sentence for the above sentence.

- Example: “Some prime number is even number”.

Domain



$\text{prime}(x)$: x is prime number
 $\text{even}(x)$: x is even number

Find FOL sentence for the above sentence.

3. Nested Quantification

- “For all x and all y, if x is the parent of y then y is a child of x”

$$\forall_{x,y} \textit{Parent}(x,y) \Rightarrow \textit{Child}(y,x)$$

- “Everybody loves somebody”

$$\forall_x \exists_y \textit{Loves}(x,y)$$

The order of quantification can change the meaning of the sentence.

- “There is someone who is loved by everybody”

$$\exists_y \forall_x \textit{Loves}(x,y)$$

- “Everybody is loved by somebody”

- “Somebody loves everybody”

- “Nobody loves everyone”

Exercise 1

- Give FOL sentence for

“Gold and silver ornaments are precious”

Let $G(x)$: x is a gold ornament,
 $S(x)$: x is a silver ornament,
 $P(x)$: x is precious.

- A) $\forall x (P(x) \Rightarrow G(x) \wedge S(x))$
- B) $\forall x (G(x) \wedge S(x) \Rightarrow P(x))$
- C) $\exists x (G(x) \wedge S(x) \Rightarrow P(x))$
- D) $\forall x (G(x) \vee S(x) \Rightarrow P(x))$

Exercise 2

- Find FOL for the following sentence

“Every teacher is liked by some students”

A) $\forall_x \textit{teacher}(x) \Rightarrow \exists_y(\textit{student}(y) \Rightarrow \textit{Likes}(y, x))$

B) $\forall_x \textit{teacher}(x) \Rightarrow \exists_y(\textit{student}(y) \wedge \textit{Likes}(y, x))$

C) $\exists_y \forall_x \textit{teacher}(x) \Rightarrow (\textit{student}(y) \wedge \textit{Likes}(y, x))$

D) $\forall_x (\textit{teacher}(x) \wedge \exists_y[\textit{student}(y) \Rightarrow \textit{Likes}(y, x)])$

Exercise 3

- Find FOL for the following sentence

“Some boys are taller than all girls.”

A) $\exists_x \textit{boy}(x) \Rightarrow \forall_y(\textit{girl}(y) \wedge \textit{taller}(x, y))$

B) $\exists_x \textit{boy}(x) \wedge \forall_y(\textit{girl}(y) \wedge \textit{taller}(x, y))$

C) $\exists_x \textit{boy}(x) \Rightarrow \forall_y(\textit{girl}(y) \Rightarrow \textit{taller}(x, y))$

D) $\exists_x \textit{boy}(x) \wedge \forall_y[\textit{girl}(y) \Rightarrow \textit{taller}(x, y)]$

Exercise 4

- Let $F(x,y,t)$: person x can fool person y at time t .

$$\forall x \exists y \exists t (\neg F(x, y, t))$$

A) Everyone can fool some person at some time.

B) No one can fool everyone all the time.

$$\neg \exists x \forall y \forall t F(x, y, t)$$

C) Everyone can not fool some person all the time.

D) No person can fool some person at some time.

Exercise 5

- Find FOL sentence for

“Everyone who loves all animal is loved by someone.”

A) $\forall_{x,y} [\textit{Animal}(y) \Rightarrow \textit{Loves}(x,y)] \Rightarrow \exists_z \textit{Loves}(z,x)$

B) $\forall_{x,y} [\textit{Animal}(y) \wedge \textit{Loves}(x,y)] \Rightarrow \exists_z \textit{Loves}(z,x)$

C) $\forall_{x,y} [\textit{Animal}(y) \Rightarrow \textit{Loves}(x,y)] \wedge \exists_z \textit{Loves}(z,x)$

Connections between $\forall$ and $\exists$

Two quantifiers are connected through negation.

$\forall x \neg Likes(x, Parsnips)$ is equivalent to $\neg \exists x Likes(x, Parsnips)$.

$\forall x Likes(x, IceCream)$ is equivalent to $\neg \exists x \neg Likes(x, IceCream)$.

The De Morgan rules for quantified and unquantified sentences are as follows:

$$\forall x \neg P \equiv \neg \exists x P$$

$$\neg \forall x P \equiv \exists x \neg P$$

$$\forall x P \equiv \neg \exists x \neg P$$

$$\exists x P \equiv \neg \forall x \neg P$$

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$P \wedge Q \equiv \neg(\neg P \vee \neg Q)$$

$$P \vee Q \equiv \neg(\neg P \wedge \neg Q).$$

Equity

We can use the equality symbol to signify that two terms refer to the same object.

Example: Father (John)=Henry

We can also use equality for counting in first-order logic

- “Spot has at least two sisters”

$$\exists_{x,y} \textit{Sister}(\textit{Spot}, x) \wedge \textit{Sister}(\textit{Spot}, y) \wedge \neg(x = y)$$

Kinship domain

Kinship domain represents the relationship between objects.

- Domain mother

$$\forall_{m,c} \textit{Mother}(c) = m \Leftrightarrow \textit{Female}(m) \wedge \textit{Parent}(m, c)$$

First-order logic for Wumpus World

We will take a look at how first-order logic reduces the rules in a Wumpus world.

Rules : *Percept* $\Rightarrow$ *Action*

Example:

“If the agent senses a glitter, it should do a grab in order to pick up the gold” can be represented by the sentence in first-order logic as follow:

$$\forall_{s,b,u,c,t} \text{Percept}([s, b, \text{Glitter}, u, c], t) \Rightarrow \text{Action}(\text{Grab}, t)$$

We can reduce some facts into a shorter representation and then use it in the following rules.

$$\forall_{s,b,u,c,t} \text{Percept}([s, b, \text{Glitter}, u, c], t) \Rightarrow \text{At}_{\text{Gold}}(t) \quad (\text{Causal Rules})$$

$$\forall_t \text{At}_{\text{Gold}}(t) \Rightarrow \text{Action}(\text{Grab}, t) \quad (\text{Diagnostic Rules})$$